

This key should allow you to understand why you choose the option you did (beyond just getting a question right or wrong). More instructions on how to use this key can be found [here](#).

If you have a suggestion to make the keys better, please fill out the short survey [here](#).

Note: This key is auto-generated and may contain issues and/or errors. The keys are reviewed after each exam to ensure grading is done accurately. If there are issues (like duplicate options), they are noted in the offline gradebook. The keys are a work-in-progress to give students as many resources to improve as possible.

1. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$\frac{-45 - 33i}{8 - i}$$

The solution is $-5.03 - 4.75i$, which is option B.

A. $a \in [-5.6, -4.35]$ and $b \in [-6, -4.5]$

$-5.03 - 4.75i$, which is the correct option.

B. $a \in [-5.75, -5.4]$ and $b \in [32.5, 33.5]$

$-5.62 + 33.00i$, which corresponds to just dividing the first term by the first term and the second by the second.

C. $a \in [-6.2, -5.7]$ and $b \in [-4, -3]$

$-6.05 - 3.37i$, which corresponds to forgetting to multiply the conjugate by the numerator and not computing the conjugate correctly.

D. $a \in [-5.6, -4.35]$ and $b \in [-309.5, -308.5]$

$-5.03 - 309.00i$, which corresponds to forgetting to multiply the conjugate by the numerator.

E. $a \in [-327.55, -326.95]$ and $b \in [-6, -4.5]$

$-327.00 - 4.75i$, which corresponds to forgetting to multiply the conjugate by the numerator and using a plus instead of a minus in the denominator.

General Comment: Multiply the numerator and denominator by the *conjugate* of the denominator, then simplify. For example, if we have $2 + 3i$, the conjugate is $2 - 3i$.

2. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$(10 - 4i)(3 + 2i)$$

The solution is $38.00 + 8.00i$, which is option C.

A. $a \in [32, 40]$ and $b \in [7, 11]$

$38.00 + 8.00i$, which is the correct option.

B. $a \in [19, 23]$ and $b \in [25, 39]$

$22.00 + 32.00i$, which corresponds to adding a minus sign in the first term.

C. $a \in [19, 23]$ and $b \in [-33, -29]$

$22.00 - 32.00i$, which corresponds to adding a minus sign in the second term.

D. $a \in [32, 40]$ and $b \in [-11, -4]$

$38.00 - 8.00i$, which corresponds to adding a minus sign to both term.

E. $a \in [25, 35]$ and $b \in [-11, -4]$

$30.00 - 8.00i$, which corresponds to just multiplying the real terms to get the real part of the solution and the coefficients in the complex terms to get the complex part.

General Comment: You can treat i as a variable and distribute. Just remember that $i^2 = -1$, so you can continue to reduce after you distribute.

3. Simplify the expression below and choose the interval the simplification is contained within.

$$17 - 12 \div 16 * 13 - (15 * 10)$$

The solution is -142.75 , which is option B.

A. $[-142.75, -135.75]$

-142.75 , which is the correct option.

B. $[-135.06, -131.06]$

-133.058 , which corresponds to an Order of Operations error: not reading left-to-right for multiplication/division.

C. $[165.94, 170.94]$

166.942 , which corresponds to not distributing addition and subtraction correctly.

D. $[-79.5, -71.5]$

-77.5 , which corresponds to not distributing a negative correctly.

E. None of the above

You may have gotten this by making an unanticipated error. If you got a value that is not any of the others, please let the coordinator know so they can help you figure out what happened.

General Comment: While you may remember (or were taught) PEMDAS is done in order, it is actually done as P/E/MD/AS. When we are at MD or AS, we read left to right.

4. Choose the **smallest** set of Real numbers that the number below belongs to.

$$-\sqrt{\frac{64}{289}}$$

The solution is Rational, which is option D.

A. Whole

These are the counting numbers with 0 (0, 1, 2, 3, ...)

B. Integer

These are the negative and positive counting numbers (... , -3, -2, -1, 0, 1, 2, 3, ...)

C. Rational

* This is the correct option!

D. Irrational

These cannot be written as a fraction of Integers.

E. Not a Real Number

These are Nonreal Complex numbers **OR** things that are not numbers (e.g., dividing by 0).

General Comment: First, you **NEED** to simplify the expression. This question simplifies to $-\frac{8}{17}$.

Be sure you look at the simplified fraction and not just the decimal expansion. Numbers such as 13, 17, and 19 provide **long but repeating/terminating decimal expansions!**

The only ways to *not* be a Real number are: dividing by 0 or taking the square root of a negative number.

Irrational numbers are more than just square root of 3: adding or subtracting values from square root of 3 is also irrational.

5. Choose the **smallest** set of Complex numbers that the number below belongs to.

$$\frac{-13}{0} + \sqrt{187}i$$

The solution is Not a Complex Number, which is option B.

A. Rational

These are numbers that can be written as fraction of Integers (e.g., $-2/3 + 5$)

B. Irrational

These cannot be written as a fraction of Integers. Remember: π is not an Integer!

C. Nonreal Complex

This is a Complex number ($a + bi$) that is not Real (has i as part of the number).

D. Pure Imaginary

This is a Complex number ($a + bi$) that **only** has an imaginary part like $2i$.

E. Not a Complex Number

* This is the correct option!

General Comment: Be sure to simplify $i^2 = -1$. This may remove the imaginary portion for your number. If you are having trouble, you may want to look at the *Subgroups of the Real Numbers* section.

6. Solve the equation below. Then, choose the interval that contains the solution.

$$-19(-11x - 10) = -13(6x + 12)$$

The solution is -1.206 , which is option E.

A. $x \in [-1.32, -1.16]$

* $x = -1.206$, which is the correct option.

B. $x \in [-0.05, 0.13]$

$x = 0.118$, which corresponds to not distributing the negative in front of the first parentheses correctly.

C. $x \in [-0.25, -0.05]$

$x = -0.118$, which corresponds to not distributing the negative in front of the second parentheses correctly.

D. $x \in [-0.36, -0.25]$

$x = -0.26$, which corresponds to getting the negative of the actual solution.

E. There are no Real solutions.

There are no Real solutions., which corresponds to students thinking a fraction means there is no solution to the equation.

General Comment: The most common mistake on this question is to not distribute the negative in front of the second fraction correctly. The best way to avoid this is putting the numerator in parentheses, which will help you remember to distribute the negative correctly.

7. Solve the linear equation below. Then, choose the interval that contains the solution.

$$\frac{8x - 8}{7} - \frac{3x + 8}{4} = \frac{-4x - 3}{3}$$

The solution is 1.241, which is option A.

A. $x \in [0.75, 1.3]$

* $x = 1.241$, which is the correct option.

B. $x \in [-1.8, -0.95]$

$x = -1.076$, which corresponds to not distributing the negative in front of the second fraction.

C. $x \in [7.35, 7.8]$

$x = 7.531$, which corresponds to dividing the coefficients in front of x by the denominator rather than dividing BOTH parts of the numerator by the denominator (or removing the fractions through multiplication).

D. $x \in [-0.15, 0.7]$

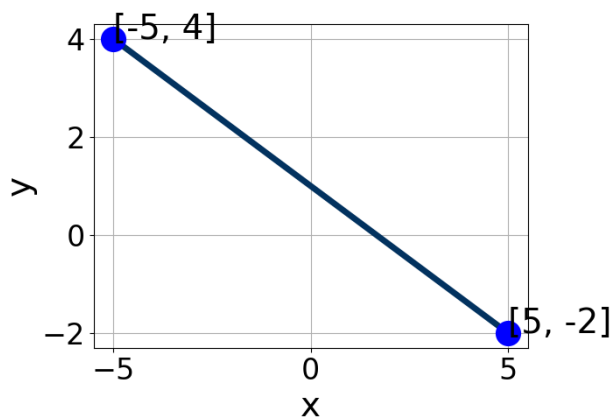
$x = 0.238$, which corresponds to dividing the second number in the numerator by the denominator rather than dividing BOTH parts of the numerator by the denominator (or removing the fractions through multiplication).

E. There are no Real solutions.

There are no Real solutions., which corresponds to students thinking a fraction means there is no solution to the equation.

General Comment: If you are having trouble with this problem, try to remove a fraction at a time by multiplying each term by the denominator.

8. Write the equation of the line in the graph below in Standard Form $Ax + By = C$. Then, choose the intervals that contain A , B , and C .



The solution is $3x + 5y = 5$, which is option E.

A. $A \in [2.5, 4.5]$, $B \in [3, 6]$, and $C \in [4.5, 6.5]$

* $3x + 5y = 5$, which is the correct option.

B. $A \in [-4, -2.5]$, $B \in [-6, -4.5]$, and $C \in [-5.5, -4.5]$

$-3x - 5y = -5$, which corresponds to not making A positive (by multiplying the equation by -1).

C. $A \in [2.5, 4.5]$, $B \in [-6, -4.5]$, and $C \in [-5.5, -4.5]$

$3x - 5y = -5$, which corresponds to using the opposite (negative) slope of the graph, but did everything else correctly.

D. $A \in [0, 2]$, $B \in [0.5, 2]$, and $C \in [-0.5, 1.5]$

$0.6x + 1y = 1.0$, which corresponds to not removing rational values for Standard Form.

E. $A \in [0, 2]$, $B \in [-1.5, 0]$, and $C \in [-2, 0]$

$0.6x - 1y = -1.0$, which corresponds to using the opposite (negative) slope of the graph and not removing rational values.

General Comment: Standard form is supposed to have $A > 0$ and all fractions removed.

9. First, find the equation of the line containing the two points below. Then, write the equation in the form $y = mx + b$ and choose the intervals that contain m and b .

$(-3, 10)$ and $(-8, -7)$

The solution is $y = 3.4x + 20.2$, which is option C.

A. $a \in [2.5, 5]$ and $b \in [19.5, 21]$

* $y = 3.4x + 20.2$, which is the correct option.

B. $a \in [-4, -3]$ and $b \in [-35, -33]$

$y = -3.4x - 34.2$, which corresponds to using the negative slope and the correct equation.

C. $a \in [2.5, 5]$ and $b \in [-22, -19.5]$

$y = 3.4x - 20.2$, which corresponds to using the correct slope and getting the negative y-intercept.

D. $a \in [2.5, 5]$ and $b \in [11.5, 13.5]$

$y = 3.4x + 13$, which corresponds to using the correct slope/equation but not distributing correctly using the first point.

E. $a \in [2.5, 5]$ and $b \in [0.5, 2.5]$

$y = 3.4x + 1$, which corresponds to using the correct slope/equation but not distributing correctly using the second point.

General Comment: Remember to keep your points in order when plugging in to the slope formula.

10. Find the equation of the line described below. Write the linear equation in the form $y = mx + b$ and choose the intervals that contain m and b .

Perpendicular to $9x - 5y = 7$ and passing through the point $(7, -5)$.

The solution is $y = -0.56x - 1.11$, which is option E.

A. $a \in [-1.5, 0]$ and $b \in [-2, -1]$

* $y = -0.56x - 1.11$, which is the correct option.

B. $a \in [-0.5, 2]$ and $b \in [-9.5, -8.5]$

$y = 0.56x - 8.89$, which corresponds to using the negative slope.

C. $a \in [-1.5, 0]$ and $b \in [-0.5, 2]$

$y = -0.56x + 1.11$, which corresponds to using the correct slope and getting the negative y -intercept.

D. $a \in [-1.5, 0]$ and $b \in [-13, -11]$

$y = -0.56x - 12.00$, which corresponds to correct slope and mis-distributing while simplifying to slope-intercept form.

E. $a \in [-2.5, -1]$ and $b \in [-2, -1]$

$y = -1.80x - 1.11$, which corresponds to using the reciprocal slope ($1/m$).

General Comment: Parallel slope is the same and perpendicular slope is opposite reciprocal. Opposite reciprocal means flipping the fraction and changing the sign (positive to negative or negative to positive).

11. Using an interval or intervals, choose the option that describes all the x -values within or including a distance of the given values.

Less than 6 units from the number 4.

The solution is $(-2, 10)$, which is option C.

A. $[-2, 10]$

This describes the values no more than 6 from 4

B. $(-2, 10)$

* This is the correct option!

C. $(-\infty, -2) \cup (10, \infty)$

This describes the values more than 6 from 4

D. $(-\infty, -2] \cup [10, \infty)$

This describes the values no less than 6 from 4

E. None of the above

You likely thought the values in the interval were not correct.

General Comment: When thinking about this language, it helps to draw a number line and try points.

12. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-4 - 3x \leq \frac{-12x - 3}{5} < 4 - 3x$$

The solution is None of the above., which is option D.

- A. $(a, b]$, where $a \in [3.75, 10.5]$ and $b \in [-9, -6.75]$
 $(5.666666666666667, -7.666666666666667]$, which corresponds to flipping the inequality and getting negatives of the actual endpoints.
- B. $[a, b)$, where $a \in [-1.5, 10.5]$ and $b \in [-12.75, -5.25]$
 $[5.666666666666667, -7.666666666666667)$, which is the correct interval but negatives of the actual endpoints.
- C. $(-\infty, a) \cup [b, \infty)$, where $a \in [3, 7.5]$ and $b \in [-11.25, -6]$
 $(-\infty, 5.666666666666667) \cup [-7.666666666666667, \infty)$, which corresponds to displaying the and-inequality as an or-inequality AND flipping the inequality AND getting negatives of the actual endpoints.
- D. $(-\infty, a] \cup (b, \infty)$, where $a \in [0.75, 8.25]$ and $b \in [-12, -3]$
 $(-\infty, 5.666666666666667] \cup (-7.666666666666667, \infty)$, which corresponds to displaying the and-inequality as an or-inequality and getting negatives of the actual endpoints.
- E. None of the above.
- * This is the correct option!

General Comment: To solve, you will need to break up the compound inequality into two inequalities. Be sure to keep track of the inequality! It may be best to draw a number line and graph your solution.

13. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-5 + 4x > 5x \text{ or } -3 + 9x < 10x$$

The solution is $(-\infty, -5.0)$ or $(-3.0, \infty)$, which is option E.

- A. $(-\infty, a)$ or (b, ∞) , where $a \in [-6, -3.5]$ and $b \in [-4, -2.5]$
 * $(-\infty, -5.0)$ or $(-3.0, \infty)$, which is the correct option.
- B. $(-\infty, a)$ or (b, ∞) , where $a \in [1.5, 3.5]$ and $b \in [-5.5, -4]$
 $(-\infty, 3.0)$ or $(5.0, \infty)$, which corresponds to inverting the inequality and negating the solution.
- C. $(-\infty, a]$ or $[b, \infty)$, where $a \in [-5.5, -3.5]$ and $b \in [-3.5, -2.5]$
 $(-\infty, -5.0]$ or $[-3.0, \infty)$, which corresponds to including the endpoints (when they should be excluded).
- D. $(-\infty, a]$ or $[b, \infty)$, where $a \in [2, 4]$ and $b \in [-6.5, -4.5]$
 $(-\infty, 3.0]$ or $[5.0, \infty)$, which corresponds to including the endpoints (when they should be excluded) and inverting the inequality and negating the solution.
- E. $(-\infty, \infty)$
 $(-\infty, \infty)$, which corresponds to the variable canceling and does not happen in this instance.

General Comment: When multiplying or dividing by a negative, flip the sign.

14. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-6x + 9 \leq 10x + 3$$

The solution is $[0.375, \infty)$, which is option A.

A. $[a, \infty)$, where $a \in [0.2, 1.3]$

* $[0.375, \infty)$, which is the correct option.

B. $(-\infty, a]$, where $a \in [-0.2, 0.61]$

$(-\infty, 0.375]$, which corresponds to switching the direction of the interval. You likely did this if you did not flip the inequality when dividing by a negative!

C. $[a, \infty)$, where $a \in [-2, 0.2]$

$[-0.375, \infty)$, which corresponds to negating the endpoint of the solution.

D. $(-\infty, a]$, where $a \in [-0.84, -0.16]$

$(-\infty, -0.375]$, which corresponds to switching the direction of the interval AND negating the endpoint. You likely did this if you did not flip the inequality when dividing by a negative as well as not moving values over to a side properly.

E. None of the above

None of the above. You may have chosen this if you thought the inequality did not match the ends of the intervals.

General Comment: Remember that less/greater than or equal to includes the endpoint, while less/greater do not. Also, remember that you need to flip the inequality when you multiply or divide by a negative.

15. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$\frac{3}{3} - \frac{3}{6}x < \frac{6}{5}x - \frac{6}{7}$$

The solution is $(1.092, \infty)$, which is option E.

A. (a, ∞) , where $a \in [-0.3, 5.2]$

* $(1.092, \infty)$, which is the correct option.

B. $(-\infty, a)$, where $a \in [-0.91, 5.09]$

$(-\infty, 1.092)$, which corresponds to switching the direction of the interval. You likely did this if you did not flip the inequality when dividing by a negative!

C. (a, ∞) , where $a \in [-3.3, -0.5]$

$(-1.092, \infty)$, which corresponds to negating the endpoint of the solution.

D. $(-\infty, a)$, where $a \in [-3.09, -0.09]$

$(-\infty, -1.092)$, which corresponds to switching the direction of the interval AND negating the endpoint. You likely did this if you did not flip the inequality when dividing by a negative as well as not moving values over to a side properly.

E. None of the above

None of the above. You may have chosen this if you thought the inequality did not match the ends of the intervals.

General Comment: Remember that less/greater than or equal to includes the endpoint, while less/greater do not. Also, remember that you need to flip the inequality when you multiply or divide by a negative.

16. Solve the quadratic equation below. Then, choose the intervals that the solutions x_1 and x_2 belong to, with $x_1 \leq x_2$.

$$15x^2 + 38x + 24$$

The solution is $x_1 = -1.333$ and $x_2 = -1.200$, which is option A.

- A. x_1 in $[-1.6, 1.01]$ and x_2 in $[-1.22, -1.16]$

* $x_1 = -1.333$ and $x_2 = -1.200$, which is the correct option. Obtained by solving the factored version $(3x + 4)(5x + 6)$.

- B. x_1 in $[-20.64, -19.3]$ and x_2 in $[-18.02, -17.97]$

$x_1 = -20.000$ and $x_2 = -18.000$, which corresponds to solving the factored version $(x + 20)(x + 18)$.

- C. x_1 in $[-6.85, -4.48]$ and x_2 in $[-0.29, -0.05]$

$x_1 = -6.000$ and $x_2 = -0.267$, which corresponds to solving the factored version $(x + 6)(15x + 4)$.

- D. x_1 in $[-3.24, -1.62]$ and x_2 in $[-0.75, -0.6]$

$x_1 = -2.400$ and $x_2 = -0.667$, which corresponds to solving the factored version $(5x + 12)(3x + 2)$.

- E. x_1 in $[-5.12, -3.68]$ and x_2 in $[-0.55, -0.32]$

$x_1 = -4.000$ and $x_2 = -0.400$, which corresponds to solving the factored version $(3x + 12)(5x + 2)$.

General Comment: This question can be factored, but it may be faster to find the solutions via the Quadratic Equation.

17. Factor the quadratic below. Then, choose the intervals that contain the constants in the form $(ax + b)(cx + d)$; $b \leq d$.

$$24x^2 + 2x - 35$$

The solution is $(6x - 7)(4x + 5)$, which is option A.

- A. $a \in [5.32, 8.33]$, $b \in [-17, -6]$, $c \in [3.45, 4.01]$, and $d \in [-5, 7]$

* $(6x - 7)(4x + 5)$, which is the correct option.

- B. $a \in [0.45, 1.82]$, $b \in [-31, -23]$, $c \in [0.9, 1.34]$, and $d \in [27, 32]$

$(x - 28)(x + 30)$, which corresponds to factoring $x^2 + 2x - 840$.

- C. $a \in [2.37, 3.15]$, $b \in [-17, -6]$, $c \in [7.84, 8.28]$, and $d \in [-5, 7]$

$(3x - 7)(8x + 5)$, which corresponds to associating some factor of c to a .

- D. $a \in [11.39, 12.09]$, $b \in [-17, -6]$, $c \in [1.49, 2.25]$, and $d \in [-5, 7]$

$(12x - 7)(2x + 5)$, which corresponds to associating some factor of a to c .

- E. None of the above.

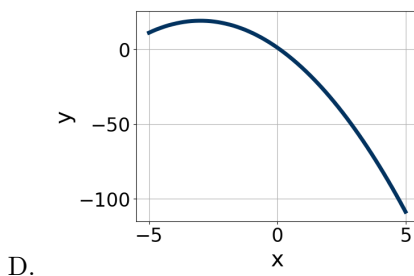
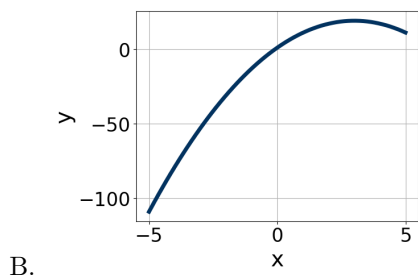
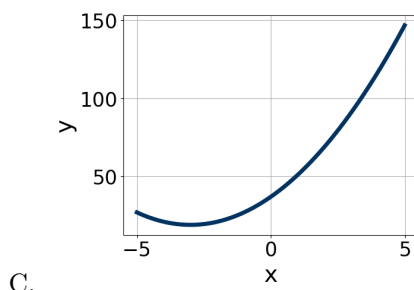
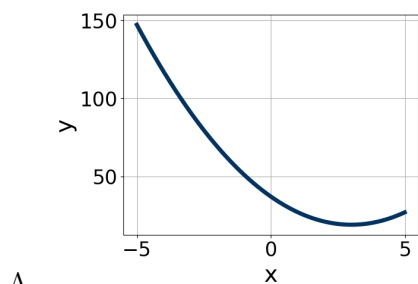
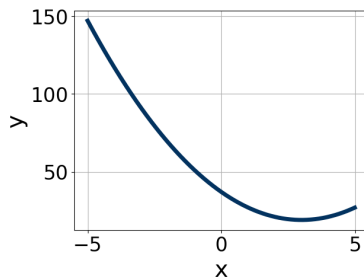
None of the above, which corresponds to a different factoring than any of the predicted options. If you get this, please let the coordinator know so they can work with you to figure out what went wrong with your factoring.

General Comment: ac had many factors in this problem. It is best to list out the possible pairs in order to make sure you don't miss any.

18. Choose the graph of the equation below.

$$f(x) = (x - 3)^2 + 19$$

The solution is the graph below, which is option A.



E. None of the above.

General Comment: Remember that Vertex Form is $y = a(x - h)^2 + k$, where the vertex is (h, k) .

19. Solve the quadratic equation below. Then, choose the intervals that the solutions belong to, with $x_1 \leq x_2$ (if they exist).

$$-13x^2 - 7x + 2$$

The solution is $x_1 = -0.745$ and $x_2 = 0.207$, which is option B.

A. x_1 in $[-0.87, -0.45]$ and x_2 in $[-0.94, 0.58]$

* $x_1 = -0.745$ and $x_2 = 0.207$, which is the correct option.

B. x_1 in $[0.22, 0.78]$ and x_2 in $[5.96, 6.97]$

$x_1 = 0.298$ and $x_2 = 6.702$, which corresponds to using the Quadratic Formula with $a = 1$.

C. x_1 in $[-0.5, -0.09]$ and x_2 in $[0.61, 0.79]$

$x_1 = -0.207$ and $x_2 = 0.745$, which corresponds to using the Quadratic Formula as $\frac{b \pm \sqrt{b^2 - 4ac}}{2a}$

D. x_1 in $[-12.77, -12.62]$ and x_2 in $[11.45, 12.42]$

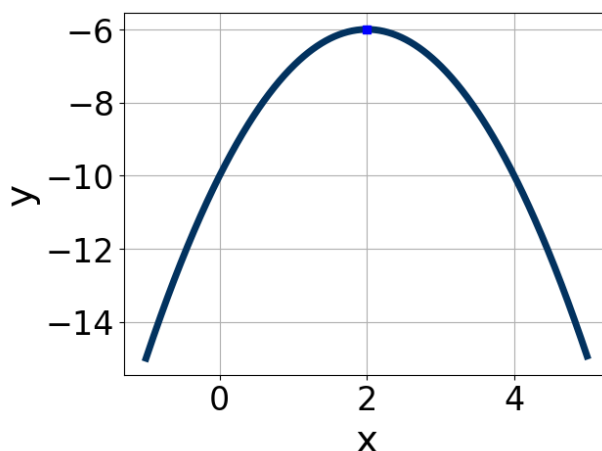
$x_1 = -12.639$ and $x_2 = 12.100$, which corresponds to using the Quadratic Formula as $-\frac{b}{2a} \pm \sqrt{b^2 - 4ac}$.

E. There are no Real solutions.

Corresponds to getting a negative under the radical or believing that since the quadratic cannot be factored, it has no Real solutions.

General Comment: This requires Quadratic Formula. Just be sure to use the correct formula and watch your signs.

20. Write the equation of the graph presented below in the form $f(x) = ax^2 + bx + c$, assuming $a = 1$ or $a = -1$. Then, choose the intervals that a , b , and c belong to.



The solution is $-x^2 + 4x - 10$, which is option C.

A. $a \in [-1.7, -0.2]$, $b \in [2, 6]$, and $c \in [-11, -8]$

* $-x^2 + 4x - 10$, which is the correct option.

B. $a \in [-1.7, -0.2]$, $b \in [-9, -3]$, and $c \in [-11, -8]$

$f(x) = -x^2 - 4x - 10$, which corresponds to incorrectly using vertex form as $f(x) = a(x + h)^2 + k$.

C. $a \in [0.1, 1.2]$, $b \in [2, 6]$, and $c \in [-3, -1]$

$f(x) = x^2 + 4x - 2$, which corresponds to incorrectly using vertex form as $f(x) = a(x + h)^2 + k$ AND making a the opposite sign than it should be.

D. $a \in [-1.7, -0.2]$, $b \in [-9, -3]$, and $c \in [1, 3]$

$f(x) = -x^2 - 4x + 2$, which corresponds to incorrectly using vertex form as $f(x) = a(x + h)^2 - k$.

E. $a \in [0.1, 1.2]$, $b \in [-9, -3]$, and $c \in [-3, -1]$

$f(x) = x^2 - 4x - 2$, which corresponds to making a the opposite sign than it should be.

General Comment: When the graph is pointing up, $a = 1$. When the graph is pointing down, $a = -1$. Be sure to use Vertex Form: $y = a(x - h)^2 + k$.